

DAY – 01 EXPERIMENTS

1. CALCULATOR PROGRAM:

QUESTIONS 9 / 48 completed

Arithmetic

ARITHMETIC

Write a program to read in two integers and perform the following operations on them: addition, subtraction, multiplication, division, and modulo.

PROGRAM EDITOR

C++ Run Save

```
1 #include<iostream>
2 using namespace std;
3 int main(){
4     int a,b;
5     cin>>a;
6     cin>>b;
7     cout<<"The Addition of a&b:"<<a+b<<endl;
8     cout<<"The Subtraction of a&b:"<<a-b<<endl;
9     cout<<"The Multiplication of a&b:"<<a*b<<endl;
10    cout<<"The Division of a&b:"<<a/b<<endl;
11    cout<<"The Modulo of a&b:"<<a%b;
12    return 0;
13 }
```

OUTPUT

```
The Addition of a&b:15
The Subtraction of a&b:5
The Multiplication of a&b:50
The Division of a&b:2
The Modulo of a&b:0
```

INPUT

```
10
5
```

2. FACTORIAL PROGRAM:

QUESTIONS 9 / 48 completed

Factorial

FACTORIAL

Write a program to read in an integer and print the factorial of that number using a for loop.

PROGRAM EDITOR

C++ Run Save

```
1 #include<iostream>
2 using namespace std;
3 int main(){
4     int a,i;
5     int fact=1;
6     cin>>a;
7     for(i=1;i<=a;i++){
8         fact=fact*i;
9     }
10    cout<<"The Factorial of "<<a<<": "<<fact;
11    return 0;
12 }
```

OUTPUT

```
The Factorial of 3: 6
```

INPUT

```
3
```

3. PRIME

QUESTIONS 9 / 48 completed

Prime

PRIME

Write a program to read in an integer and determine if it is a prime number or not using a for loop.

PROGRAM EDITOR

C++ Run Save

```
1 #include<iostream>
2 using namespace std;
3 int main(){
4     int a,i;
5     cin>>a;
6     if(a>=1){
7         for(i=2;i<a;i++){
8             if(a%i==0){
9                 cout<<a<<" is Not Prime";
10                return 0;
11            }
12        }
13        cout<<a<<" is Prime";
14    }
15    else{
16        cout<<a<<" is Not Prime";
17    }
18    return 0;
19 }
```

OUTPUT

```
7 is Prime
```

INPUT

```
7
```

4. PALINDROME PROGRAM

QUESTIONS 9 / 48 completed

Palindrome

PALINDROME

Write a program to read in an integer and determine if it is a palindrome or not using a while loop.

PROGRAM EDITOR

C++ Run Save

```
1 #include<iostream>
2 using namespace std;
3 int main(){
4     int n,temp,reverse=0,rem;
5     cin>>n;
6     temp=n;
7     while(n!=0){
8         rem=n%10;
9         reverse=reverse*10+rem;
10        n=n/10;
11    }
12    if (temp==reverse)
13        cout<<temp<<" is a Palindrome";
14    else
15        cout<<temp<<" is not a Palindrome";
16    return 0;
17 }
```

OUTPUT

```
7 is a Palindrome
```

INPUT

```
7
```

5. FIBONACCI SERIES

QUESTIONS 10 / 48 completed

Fibonacci

FIBONACCI

Write a program to read in an integer and print the Fibonacci series up to that number using a for loop.

PROGRAM EDITOR

C++

Run Save

```
1 #include<iostream>
2 using namespace std;
3 int main(){
4     int n,a=0,b=1,c,i;
5     cin>>n;
6     for(i=1;i<=n;i++){
7         cout<<a<<" ";
8         c=a+b;
9         a=b;
10        b=c;
11    }
12    return 0;
13 }
```

OUTPUT

```
0 1 1 2 3 5 8
```

INPUT

7

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